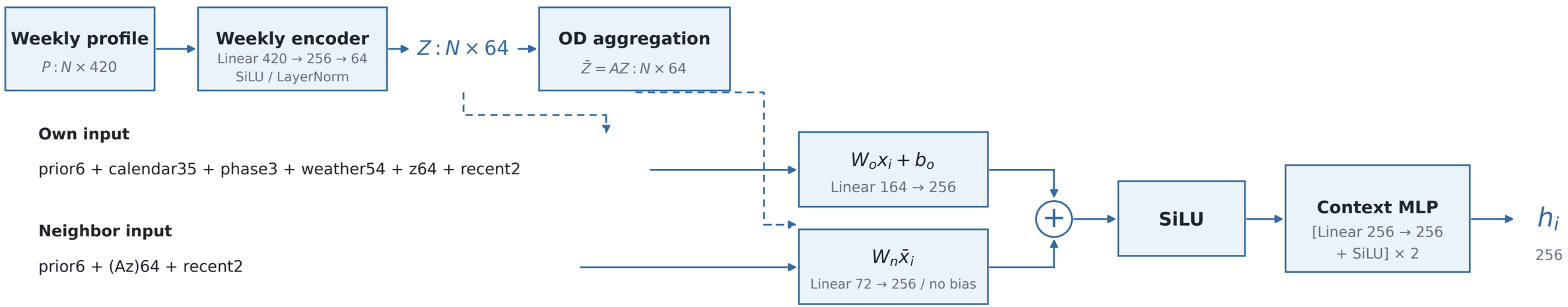
# H13 encoder and Flow Matching decoder

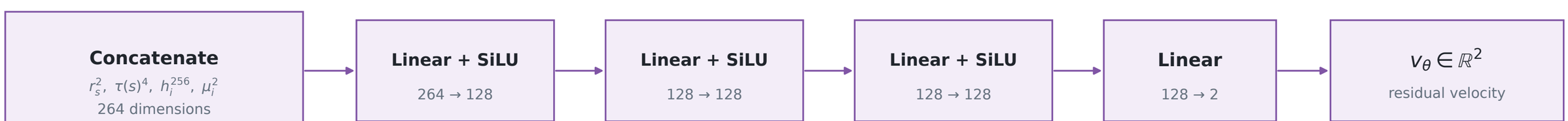
Exact layer dimensions in the current implementation

## (a) Encoder: weekly profile, fixed graph aggregation, and parallel projections




The graph operator A is fixed from historical OD records. Calendar, weather and recent-value encodings are unchanged.

## (b) Decoder: residual velocity network




$$\mu_i = \log(1 + \lambda_i), \qquad \tau(s) = [s, \sin(\pi s), \cos(\pi s), s^2]$$

**FM training** $r_1 = \log(1 + y + u) - \mu_i, \quad r_s = (1 - s) r_0 + s r_1, \quad \mathcal{L}_{\mathrm{FM}} = \| v_\theta(r_s, s \mid h_i, \mu_i) - (r_1 - r_0) \|^2$

$r_0 \sim \mathcal{N}(0, I_2), \quad u \sim U([0,1)^2), \quad s \sim U([0,1])$. The count network is frozen while training FM.

Reference implementation: forecast_transfer.py / rolling_inventory_model.py / scaled_flow_core.py (5db8c21).